# Take-Home Assignment for Segment 05

Bacteria use alternative start codons GUG and UUG (GTG and TTG in DNA) in addition to the canonical AUG (ATG). Bacterial genomes also vary significantly in their GC content (from ~20% to over 70%). One can hypothesize that bacteria with GC-rich genomes may be using GTG more often than those with AT-rich genomes. Test the hypothesis by doing the following:

1. Write a program (in C++) that reads the FASTA file with nucleotide sequences of all genes encoded in the genome and counts the number of genes that start with ATG, GTG, and TTG (if a gene starts with something else, discard it but print the list of such genes; it could be an error or a pseudogene).
2. Download files with all gene sequences for at least six bacterial genomes that span a wide range of GC content and use your program to count genes starting with the three different start codons.
3. Write a brief conclusion whether the percentage of genes starting with GTG correlates with the genome GC content and provide supporting evidence (e.g., a plot %GTG vs. genome GC content).
4. Submit your code and the brief report as separate files via eLC

Maximum award for completing the assignment is 15 points (10 for the program that reports counts and percentages of different start codons, 2 points for listing genes with unexpected start codons, and 3 points for the report that describes the results and formulates a conclusion).

Notes and useful resources:

- Sample input (H.pyloriJ99-genes.txt) and sample output using my program for this assignment (H.pyloriJ99-genes-StartCodons.txt) are attached.
- This is a list of sequenced prokaryotic genomes and their GC content (it includes bacteria and archaea – do not use archaea): [http://insilico.ehu.es/oligoweb/index2.php?m=all](http://insilico.ehu.es/oligoweb/index2.php?m=all). You can use the genome GC content from this online table for your report rather than calculating it yourself.
- You can download the input files from GenBank. Go to [https://www.ncbi.nlm.nih.gov/](https://www.ncbi.nlm.nih.gov/), select the “genome” database, and search for the name of the organism (e.g., *Myxococcus xanthus*). This should take you to the list of sequenced strains ([https://www.ncbi.nlm.nih.gov/datasets/genome/?taxon=34](https://www.ncbi.nlm.nih.gov/datasets/genome/?taxon=34)) of the species. Find the strain that you want to download. In the “Action” column, select “Download”, then select “RefSeq only” and “Genomic coding sequences (FASTA)” (uncheck the “Genome sequences” unless you want to download those, too). This will download a zip archive from which you can extract the file and rename it appropriately. It will probably be named “cds_from_genomic.fna” – it is a plain text file, so you can change the extension to .txt if you want.

- I leave it up to you how you present the data but keep in mind that the goal is to answer the question whether your results support the hypothesis that genomes with higher GC content use GTG start codons more often. If you do not know anything else, get the data for at least 6 genomes with varying GC content, make a plot of % genes starting with GTG vs genome GC content, and decide whether you think that there is a relationship. If you know how, you can determine a correlation coefficient and/or assess statistical significance, but that is not required.
- If you want to assess statistical significance of a correlation coefficient, here's one tool I sometimes use: http://vassarstats.net/rsig.html
- Note in your conclusions if you find anything surprising or unexpected.
- If the results do not agree with the hypothesis or the sample is too small to make a conclusion, don't be afraid to say that.